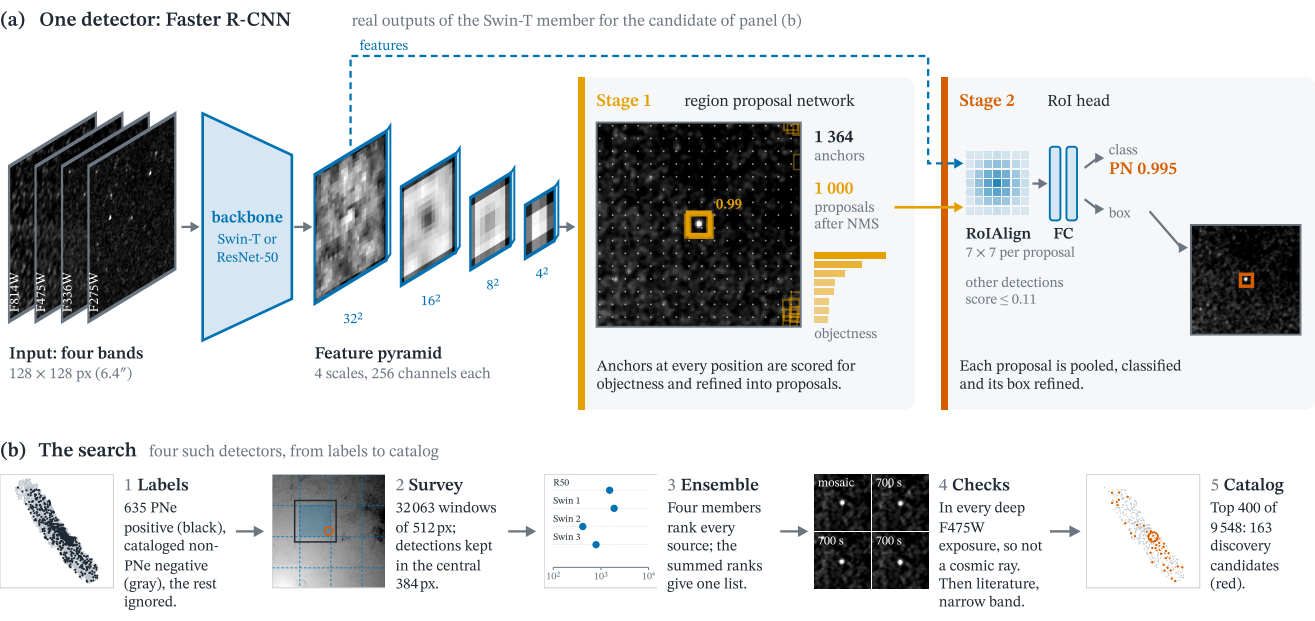

(a) One detector: Faster R-CNN
real outputs of the Swin-T member for the candidate of panel (b)
features
F814W
F475W
F336W
F225W
backbone
Swin-T or
ResNet-50
32²
16²
8²
4²
Input: four bands
128 × 128 px (6.4″)
Feature pyramid
4 scales, 256 channels each
Stage 1
region proposal network
0.99
1 364
anchors
1 000
proposals
after NMS
objectness
Anchors at every position are scored for
objectness and refined into proposals.
Stage 2
RoI head
RoIAlign
7 × 7 per proposal
FC
class
PN 0.995
box
other detections
score ≤ 0.11
Each proposal is pooled, classified
and its box refined.
(b) The search
four such detectors, from labels to catalog
1 Labels
635 PNe
positive (black),
cataloged non-
PNe negative
(gray), the rest
ignored.
2 Survey
32 063 windows
of 512 px;
detections kept
in the central
384 px.
R50
Swin 1
Swin 2
Swin 3
10²
10³
10⁴
3 Ensemble
Four members
rank every
source; the
summed ranks
give one list.
mosaic
700 s
700 s
700 s
4 Checks
In every deep
F475W
exposure, so not
a cosmic ray.
Then literature,
narrow band.
5 Catalog
Top 400 of
9 548: 163
discovery
candidates
(red).